

Requirements – Level B

Acme Starters, Inc. has requested several additional requirements that are described in this document. The new version of the project must be code named “Acme-Starters-B”.

Requirement Zero

Before any implementation work begins, every group must engage in thorough reading, reflection, analysis, planning, and preparation. These early stages are critical to the success of your project since they will help reduce rework, bottlenecks, and misunderstandings later on.

To guide your preparation, please ensure that your group undertakes the following activities:

- Rewrite the requirements in a structured form. Transform the requirements in this document into a clear, systematic specification that includes information requirements, functional requirements, non-functional requirements, testing requirements, and so on. Follow the patterns used in the requirements for Level C, with special emphasis on identifying constraints both in the data model and the features that your system must support. This rewriting will help you understand the requirements, detect inconsistencies, incompleteness, unveil hidden constraints, and so on.
- Prepare the foundations of the analysis document. Begin drafting your analysis document, even at a preliminary stage. As you work through the requirements, record all questions arising from your interpretation, any doubts or points that need clarification, and issues that must be confirmed by your lecturers. A well-structured list of questions will enable more efficient interactions and will reduce delays.
- Outline a preliminary project plan. Prepare an initial project schedule aimed at maximising useful working time and minimising dependencies amongst the members of your workgroup or the lecturers.

To help ensure solid preparation, we also strongly advise that you should:

- Ensure that every group member fully understands the scope and goals of Level B before any other work begins.
- Ensure that every group member who has passed level C can commit to working on Level B. Review and update your charter if necessary.
- Ensure that every group member understands the methodology taught for working together. Refrain from working on this project if you are not willing to follow it.
- Conduct a risk analysis to identify potential technical and organisational risks and propose sensible mitigation strategies. Review and update your charter if necessary.
- Establish a shared workspace in which you all can work according to the methodology taught.

Requirements

Information requirements

B. Model implementation

The workgroup must implement the constraints that are sketched in file "Model-B.uxf".

Functional requirements

B0. Any authenticated principal must be able to:

1. Acquire role manager as long as he or she does not have that role yet, and update the corresponding role profile.

B1. Any principal must be able to:

1. List and display the projects in the system as long as they have been published. ·
The principal must be able to navigate to their components and display them. He or she must be able to display the profile of the corresponding manager and list and display the Inventors, SpokesPeople, and Fundraisers that are members of the project.

B2. An principal with role manager must be able to:

1. Manage his or her projects, which involves creating, listing and displaying them.
2. Manage the team of his or her projects by selecting members among the following roles: inventors, spokespeople, and fundraisers. He or she must be able to list and display inventors, spokespeople and fundraisers that aren't in that project yet and have the possibility to add them to the project.
3. Publish any of his or her projects. This action requires a minimum of one invention to be associated with the project. Note that publishing a project implies that its components (inventions, campaigns, and strategies) must be published automatically.
4. Display a dashboard with the following indicators:
 - The total number of projects that he or she manages.
 - The deviation of that total number with respect to the average number of projects managed by the other managers.
 - The minimum, the maximum, the average, and the deviation of the effort required by his or her projects.

B3. A project member must be able to:

1. Add their inventions, campaigns, and strategies to the projects in which they are involved, even if such components are not published, yet.
2. Display all the data involved in the projects in which they are involved, even if they are not published, yet.

B4. A principal with role sponsor or auditor must be able to:

1. Sponsorship or audit, respectively, any published projects by attaching any of their, respectively, published sponsorships or audit reports.

B5. An administrator must be able to:

1. Manage the ad banners in the system, which involves creating, updating, and deleting them.
2. Store and maintain the following data for each banner: a slogan, a target URL, and a picture.

Non-functional requirements

B6. Business logic regarding effort:

- The effort of a project must be computed in PMs ("person-months") as the sum of active months of its components divided by the number of people involved.

B7. Usability:

- Ad banners must be shown as little intrusively as possible.

B8. Internationalisation:

- The interface must be available in English and Spanish. Boolean, moment-related, money-related, and numeric data must be properly internationalised.

Testing requirements

B9. Sample data:

- Produce sample data to test all the requirements above.

B10. Error data:

- Produce error data for each entity to ensure data integrity.

Managerial requirements

Intentionally blank.

Audit requirements

B11. The customer's auditor requires the following documents:

- A planning report, an analysis report (including a UML data model), a Lint report, and a testing report.